Unit-II

Chapter

2

Next Generation IP

Syllabus :

IPv6 Addressing : Representation, Address space, Address space allocation, Autoconfiguration, Renumbering. Transition from IPv4 to IPv6 : Dual stack, Tunneling, Header translation. IPv6 protocol : Packet format, Extension header.

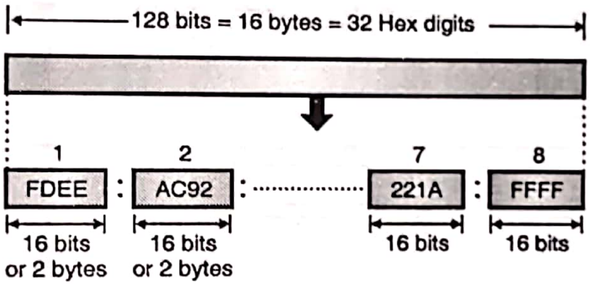
Chapter Contents

2.1 IPv6 Addressing :

- IPv6 is the next generation Internet Protocol designed as the next step of the IP version 4. IPv6 was designed to enable high-performance and larger address space. This was achieved by overcoming many of the weaknesses of IPv4 protocol and by adding several new features.
- The IPv6 was developed due to the address depletion of IPv4.
- The structure of IPv6 address is fundamentally different than that of IPv4. Therefore there is absolutely no possibility of address depletion taking place in future.

2.1.1 IPv6 Address :

- An IPv6 address is 128 bit long. It consists of 16 bytes as shown in Fig. 2.1.1. Thus the IPv6 address is 4 times longer than that of IPv4.



(G-545) **Fig. 2.1.1 : IPv6 address**

2.1.2 Notations :

- An address is stored in the computers in the binary form. But it is impossible for humans to handle a 128 bit binary address.
- Therefore many notations have been proposed to represent the IPv6 addresses, so that they become easier to handle for human beings.
- Some of the proposed notations are :
  1. Dotted decimal notation.
  2. Colon hexadecimal notation.
  3. Mixed representation.
  4. CIDR notation.

1. Dotted decimal notation :

- In order to maintain the compatibility with IPv4 addresses. We may feel tempted to use the dotted decimal notation.
- But practical observation is that this notation is convenient only for the 4 byte address of IPv4. It is not at all convenient for the 16 byte IPv6 addresses as it seems too long.
- Therefore this notation is very rarely used.

2. Colon hexadecimal notation :

- The 128 bit address can be made more readable and easy to handle. IPv6 has specified the **colon hexadecimal notation.**
- IPv6 uses a special notation called hexadecimal colon notation. In this, the total 128 bits are divided into 8 sections, each one is 16 bits or 2 bytes long.
- The 16 bits or 2 bytes in binary correspond to four hexadecimal digits of 4-bits each. Hence the 128 bits in hexadecimal form will have 8 × 4 = 32 hexadecimal digits. These are in groups of 4 digits as shown and every group is separated by a colon as shown in Fig. 2.1.2.

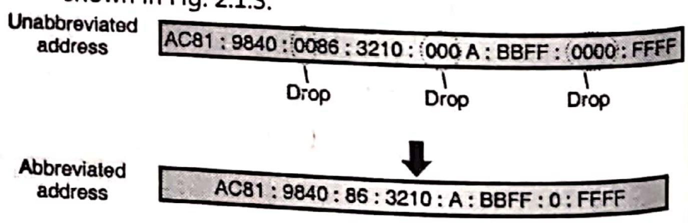
AC 81 : 9840 : 0086 : 3210 : 000A : BBFF : 0000 : FFFF

Fig. 2.1.2 : Colon hexadecimal notation

- IPv6 uses 128-bit addresses. Only about 15% of the address space is initially allocated, the remaining 85% being reserved for future use.
- These unused addresses may be used in the future for expanding the address spaces of existing address types or for totally new uses.

2.1.3 Abbreviation :

- The IPv6 address, in hexadecimal format contains 32 digits and it is very long. But in this address many hex digits are zero.
- We can take advantage of this to shorten the address by abbreviating it. A section corresponds to four digits between any two colons. The leading zeros in a section can be omitted to reduce the length of the address as shown in Fig. 2.1.3.



(G-546) **Fig. 2.1.3 : Abbreviated address**

- Note that only the leading zeros can be dropped but the trailing zeros can not be dropped. This is illustrated in Fig. 2.1.3. Thus due to abbreviation the length of the address has reduced to 24 hex digits from 32.

Further abbreviation :

- We can make further abbreviation if there are consecutive sections consisting of only zeros. This is known as **zero compression.**
- We can remove the zeros completely and replace them with double colon as shown in Fig. 2.1.4.

Abbreviated address | AC81 : 0 : 0 : 0 : 0 : BBFF : 0 : FFFF |

Replace by double colons

Further abbreviated | AC81 : : BBFF : 0 : FFFF |

(G-547) **Fig. 2.1.4 : Further abbreviation (Zero compression)**

- This further abbreviation has reduced the address length to just 13 hex digits.
- It is important to note that abbreviation can be done only once per address. Also note that if there are two sets of zero sections, then only one of them can be abbreviated.

3. Mixed representation :

- Sometimes, the IPv6 address is represented using a mixed representation which combines the **colon hex** and **dotted decimal** notations.
- This notation is appropriate during the transition time during which an IPv4 address is being embedded in IPv6 address.
- In the mixed representation the rightmost 32 bits correspond to the IPv4 address Hence they are represented by the dotted decimal notation.
- Whereas the leftmost 96 bits (6 sections) are represented in colon hex notation.

4. CIDR notation :

- The type of addressing used in IPv6 is **hierarchical addressing.** Therefore IPv6 allows classless addressing and CIDR notation.
- Fig. 2.1.5 illustrates the CIDR address with a 60 bit prefix. It has been discussed later on in this chapter, how we can divide an IPv6 address into a prefix and a suffix.

| FDEC : 0 : 0 : 0 : 0 : BBFF : 0 : FFFF |

Original address

| FDEC : : BBFF : 0 : FFFF / 60 |

CIDR address

(G-2132) **Fig. 2.1.5 : CIDR address**

Ex. 2.1.1 : IPv6 uses 16-byte addresses. If a block of 1 million addresses is allocated every picosecond, how long will be the addresses last ?

Soln. :

1. Total number of address bits = $16 \times 8 = 128$
2. Number of addresses = $2^{128} = 3.4 \times 10^{38}$
3. One picosecond = 1×10^{-12} seconds
4. 1 million addresses = 1×10^{6} address

$\therefore$ 1 picosecond = 1×10^{6} addresses

$$\therefore x = 3.4 \times 10^{38}$$

$$\therefore x = \frac{3.4 \times 10^{38}}{1 \times 10^{6}} \times 1 \text{ picoseconds}$$

$$= 3.4 \times 10^{32} \text{ picoseconds}$$

$$= 3.4 \times 10^{20} \text{ seconds}$$

$$= 9.44 \times 10^{16} \text{ hours}$$

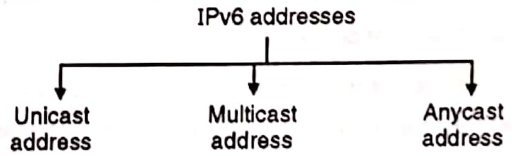
$$= 3.9352 \times 10^{15} \text{ days} = 1.0781 \times 10^{13} \text{ years}$$

2.2 Address Space :

- The address space of IPv6 contains 2^{128} addresses which is a very big number. If we compare it with the address space of IPv4, then it can be seen that, the address space of IPv6 is 2^{96} times bigger than that of IPv4.
- Therefore there is no possibility of address depletion in IPv6.

2.2.1 Three Address Types :

IPv6 defines three different types of addresses. The destination address can be one of these types :



(G-549)

1. Unicast address :

- A unicast address is meant for a single computer as a destination. A packet sent to a unicast address is meant to be delivered to the computer specified by the address.

- In IPv6 a large block of addresses has been designated from which it is possible to assign unicast addresses to the interfaces.

2. Anycast address :

- This is a type of address which is used to define a group of computers with addresses which have the same prefix.
- A packet sent to an anycast address must be delivered to only one of the member of the group which is the closest or the most easily accessible.
- No special or separate address block is assigned for anycasting in IPv6. Instead the anycast addresses are assigned from the block of unicast addresses.

3. Multicast addresses :

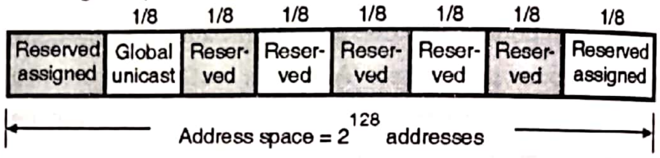
- A multicast address defines a group of computers which may or may not share the same prefix and may or may not be connected to the same physical network.
- A packet sent to a multicast address is meant to be delivered to each member of the group.
- There are no broadcast addresses in IPv6, because multicast addresses can perform the same function. The type of address is determined by the leading bits.
- All the multicast addresses start with FF (1111 1111) and all other addresses are unicast addresses.
- Anycast addresses are assigned from the unicast address space and they do not differ syntactically from unicast addresses.
- Anycast addressing is a rather new concept and there is not much experience about the widespread use of anycast addresses.
- Therefore, some restrictions apply to anycast addressing in IPv6 until more experience is gained.
- An anycast address may not be used as the Source Address of an IPv6 packet and anycast addresses may not be assigned to hosts but to routers only.
- As will be discussed later, a block is designated for multicasting in IPv6, from which the same address is assigned to the members of the group.

2.2.2 Broadcasting and Multicasting :

- In IPv6 the broadcasting is not defined at all, as in case of IPv4. In IPv6 broadcasting is considered as a special case of multicasting.

2.3 Address Space Allocation :

- Address space allocation in IPv6 is a process which divides the address space of IPv6 in several blocks. Each block is allocated for some special purpose and has a different size.
- Most of the blocks in IPv6 are not assigned yet and will be used in future.
- The entire address space in IPv6 has been divided into eight equal ranges, as shows in Fig. 2.3.1.

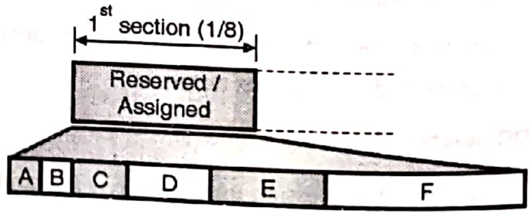


(G-2133) **Fig. 2.3.1 : Address space allocation in IPv6**

- As shown in Fig. 2.3.1, the number of addresses in each section is one eighth of the total address space. So number of addresses in each section is equal to 2^{125} addresses.

2.3.1 The First Section :

- The first section is market as Reserved/Assigned in Fig. 2.3.1. That means some address blocks in this section are reserved and the remaining are assigned as shown in Fig. 2.3.2(a).



A : 1/256 IPv4 compatible. D : 1/64 Reserved.

B : 1/256 Reserved. E : 1/32 Reserved.

C : 1/128 Reserved. F : 1/16 Reserved.

(G-2134) **Fig. 2.3.2(a) : The first section**

- As shown in Fig. 2.3.2(a), the first section is divided in six blocks of variable sizes (blocks A to F). Out of these six blocks, three blocks have been reserved and remaining three are not assigned.

2.3.2 Second Section :

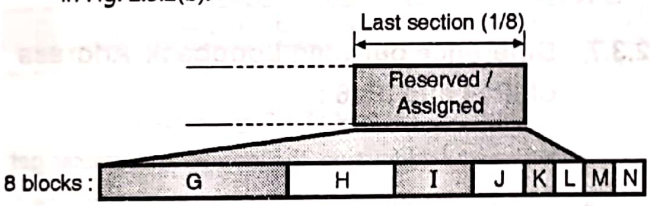
- The second section is not divided into blocks. So it is considered as one single block and is used for the **global unicast** addresses.

Sections three to seven :

These five sections from third to seventh are unassigned.

Last section :

- The last section in Fig. 2.3.2 which is marked as Reserved /Assigned is further divided into eight blocks of different sizes as shown in Fig. 2.3.2(b).
- Some of these blocks are not assigned while the other blocks are reserved for some special purpose as shown in Fig. 2.3.2(b).



G : 1/16 Reserved. K : 1/512 Reserved.

H : 1/32 Reserved. L : 1/1024 Link local

I : 1/64 Reserved. M : 1/256 Reserved.

J : 1/128 Unique local unicast N : 1/256 Multicast.

(G-2135) Fig. 2.3.2(b) : The last section

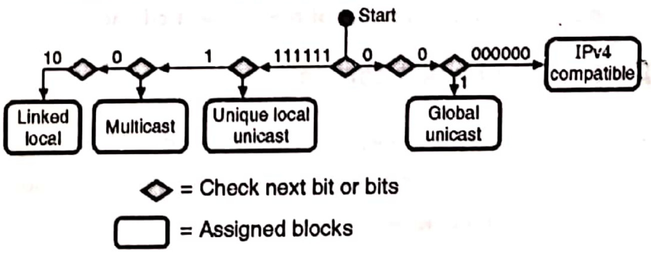
- From all this discussion it is very clear that out of the total address space of IPv6 more than 5/8th of the space is still not assigned, and only 1/8th of the address space has been assigned for the global unicast addresses for unicast communication between the users.
- The prefix for each type of address has been shown in Table 2.3.1.

Table 2.3.1 : Prefixes for IP addresses

	Block Prefix	CIDR	Block Assignment	Fraction
1	0000 0000	0000::/8	Reserved (IPv4 compatible)	1/256
	0000 0001	0100::/8	Reserved	1/256
	0000 001	0200::/7	Reserved	1/128
	0000 01	0400::/6	Reserved	1/64
	0000 1	0800::/5	Reserved	1/32
	0001	1000::/4	Reserved	1/16
2	001	2000::/3	Global unicast	1/8
3	010	4000::/3	Reserved	1/8
4	011	6000::/3	Reserved	1/8
5	100	8000::/3	Reserved	1/8
6	101	A000::/3	Reserved	1/8
7	110	C000::/3	Reserved	1/8
8	1110	E000::/4	Reserved	1/16
	1111 0	F000::/5	Reserved	1/32
	1111 10	F800::/6	Reserved	1/64
	1111 110	FC00::/7	Unique local unicat	1/128
	1111 1110 0	FE00::/9	Reserved	1/512
	1111 1110 10	FE80::/10	Link local addresses	1/1024
	1111 1110 11	FEC0::/10	Reserved	1/1024
	1111 1111	FF00::/8	Multicast addresses	1/256

2.3.3 Algorithm :

- The diagram shown in Fig. 2.3.3 has been created to illustrate that the prefixes given in Table 2.3.1 really find the block to which the IPv6 address belongs to.
- Note that, in order to make this diagram simpler we have not shown the reserved blocks.



(G-2136) Fig. 2.3.3 : An algorithm to find the allocated blocks

2.3.4 Assigned or Reserved Blocks :

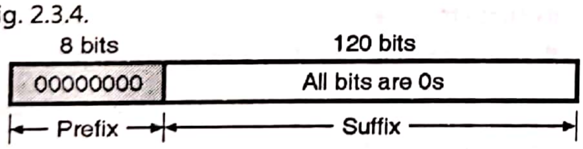
In this section, we are going to discuss the purposes and characteristics of the reserved as well as assigned blocks starting from the first row of Table 2.3.1.

IPv4 compatible addresses :

- The addresses which use the prefix (00000000) are reserved, however a part of it is used for defining some IPv4 compatible addresses.
- There are 2^{120} addresses in this block because it occupies 1/256 the fraction of the total address space.
- This block can be defined using CIDR notation as follows : 0000::/8
- This address block is further divided into many smaller subblocks. We will discuss them later in this chapter.

2.3.5 Unspecified Address :

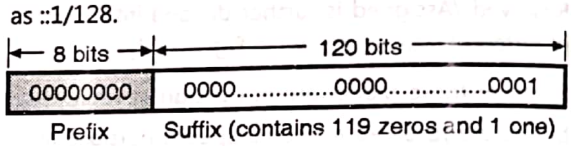
- The unspecified address is a subblock which contains only one address. By letting all suffix bits to zero, this address can be defined. That means this address is an all zeros address.
- During the bootstrapping process, when a host does not know its own address, the unspecified address is used to send an enquiry to find the unknown address of the host.
- This address is used by the host as a **source address.** However it is important to note that the unspecified address cannot be used as the **destination address.**
- This one-address subblock can be represented as ::/128 in the CIDR notation, and its format has been shown in Fig. 2.3.4.



(G-2137) **Fig. 2.3.4 : Format of the unspecified address**

2.3.6 Loopback Address :

- Like the previous one, this subblock also contains only one address which a host uses for testing itself without going into the network.
- Here the application layer creates an address, sends it to the transport layer and then passes it to the network layer.
- However it does not go to the physical layer. Instead, it returns to the transport layer and finally passed on to the application layer.
- This feature is very useful because using it we can test the functions of software package in these layers even before the computer is connected to the network.
- Fig. 2.3.5 shows the format of the loopback address which has the prefix of 00000000 and a suffix containing 119 zeros and one 1.
- We can represent this address in the CIDR notation as ::1/128.



(G-2138) **Fig. 2.3.5 : Format of the loopback address**

2.3.7 Difference between Loopback Address of IPv4 and IPv6 :

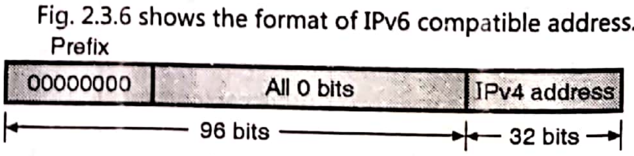
- In IPv4 classful addressing, the loop back addresses get a whole block allotted to them but in IPv6 loopback addresse is only one single address.
- In IPv4 the loop back addresses are accommodated as a part of class A address but in IPv6 it is only one single address in the reserved block.

2.3.8 Embedded IPv4 Addresses :

- As discussed in the migration from IPv4 to IPv6 in this transition period, the hosts can continue to use their IPv4 address which are embedded in IPv6 addresses.
- In order to achieve this, IPv6 has designed two formats namely **compatible format** and **mapped** format.

2.3.9 Compatible Address :

- The compatible address is an IPv6 address with 96 bits of zeros followed by 32 bits of IPv4 address.
- Fig. 2.3.6 shows the format of IPv6 compatible address.

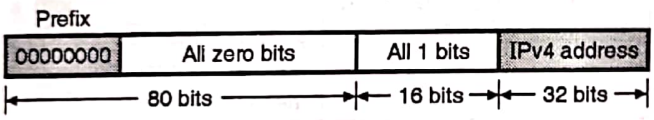


(G-2139) **Fig. 2.3.6 : Format of compatible address**

- The situation in which the compatible address is required to be used is as follows : If an IPv6 host wants to communicate with another IPv6 host but the packet is going to pass through a region in which still the IPv4 is being used by the networks.
- In order to ensure a successful passage of this packet the sender will have to use the **compatible address.**
- This is a reserved subblock which contains 2^{32} addresses and has a CIDR notation of ::/96.

2.3.10 A Mapped Address :

- The format of a mapped address is shown in Fig. 2.3.7. It shows that this address consists of 80 bits of zeros, followed by 16 bits of 1s, followed by 32 bits of IPv4 address.
- It is used when an IPv6 computer wants to communicate with an IPv4 computer.



(G-2140) **Fig. 2.3.7 : Mapped address**

- This packet can travel for an IPv6 guest, through a mostly IPv6 network and finally delivered to an IPv4 destination host.

2.3.11 Calculation of Checksum :

- The compatible and mapped addresses have been designed in such a way that the checksum can be calculated by either using the embedded address or the complete address because the extra zeros and ones are in multiples of 16. Hence they do not affect the checksum calculation in any way.
- So the value of checksum remains same even if the packet address is changed from IPv6 to IPv4.

2.4 Autoconfiguration :

- Autoconfiguration of hosts is one of the most interesting features of IPv6 addressing.
- In IPv4 the network manager originally configures the hosts and routers. Then we can allocate an IPv4 address to a host that joins the network using the DHCP protocol.
- In IPv6 we can either use DHCP protocol for allocating the IPv6 address to host or host can configure itself.
- The process of self configuration of a host when it joins the IPv6 network is as explained below.

1. The **link local address** is first created by the host for itself. This is done by following the stepwise procedure given below :
   (a) Take the 10 bit link local prefix (1111 1110 10).
   (b) Add 54 zeros to it.
   (c) Add 64 bit interface identifier which any host can generate from its interface card.

   After this we get the 128 bit link local address.
2. The host then checks for the uniqueness of this link local address. The 64 bit interface identifier also should be unique. To ensure this a neighbour solicitation message is sent by the host. Then it waits for the **neighbour advertisement message.** The process of autoconfiguration fails if any host in the subnet is using the same link local address. Then the host should use DHCP protocol to configure itself.
3. If the link local address is found to be unique, the host will store this address as its link local address for private communication. But it still requires a global unicast address. Therefore the **router solicitation message** is sent out by the host to a local router. If there is a router running on this network, then the host will receive the **router advertisement message.** This message contains the global unicast prefix and subnet prefix. The host adds these two pieces of information to its interface identifier and generates its **global unicast address.** However if it is not possible of the router to help the host with the configuration, then the router informs the host via the router advertisement message. In such a situation, the host needs to use some other means for its configuration.

2.5 Renumbering :

- In the IPv6 addressing, the facility of renumbering the address prefix (value of n) is given which allows sites to changes the service provider.
- The service provider gives a prefix number to each site to which it is connected. Therefore the site has to change the prefix number if it has to change the service provider.

- A router to which the site is connected can advertise a new prefix. The site is allowed to use its old prefix for sometime before it is finally disabled. This implies that a site uses two prefixes during the transition period.
- However the problem in using this renumbering mechanism is that it needs support of DNS.
- To overcome this problem, a new DNS protocol called **New Generation DNS** is being studied because it can support the renumbering mechanism.

2.5.1 Migrating to IPv6 (Compatibility to IPv4) :

1. It was IPv4's success that made an upgrade necessary, which means that there is a large number of IPv4 users that to be upgraded to IPv6. Keeping the transition orderly was a major objective of the entire IPng program. The cutover date when IPv6 would be turned on and IPv4 turned off has not been decided.
2. The simple strategy for upgrading involves deployment of IPv6 protocol stack in parallel with IPv4. In other words, hosts that upgrade to IPv6 will continue to simultaneously exist as IPv4 hosts.
3. An experimental IPv6 backbone, or 6 bone, has been set up to handle IPv6 Internet traffic in parallel with the regular Internet. Such hosts will continue to have 32-bit IPv4 addresses but will add 128-bit IPv6 addresses. By 1999, hundreds of networks were linked to the 6 bone.
4. The transition can be achieved through two approaches: protocol tunneling or IPv4/IPv6 dual stack.

2.5.2 Comparison between IPv4 and IPv6 :

IPv4	IPv6
In IPv4 there are only 2^{32} possible ways to represent the address (about 4 billion possible addresses)	In IPv6 there are 2^{128} possible way (about 3.4×10^{38} possible addresses)
The IPv4 address is written by dotted-decimal notation, e.g. 121.2.8.12	IPv6 is written in hexadecimal and consists of 8 groups, containing 4 hexadecimal digits or 8 groups of 16 bits each, e.g. FABC: AC77: 7834:2222:FACB: AB98: 5432:4567.
The basic length of the IPv4 header comprises a minimum of 20 bytes (without option fields). The maximum total length of the IPv4 header is 60 bytes (with option fields), and it uses 13 fields to identify various control settings.	The IPv6 header is a fixed header of 40 bytes in length, and has only 8 fields. Option information is carried by the extension header, which is placed after the IPv6 header.
IPv4 header has a checksum, which must be computed by each router	IPv6 has no header checksum because checksums are, for example, above the TCP/IP protocol suite, and above the Token Ring, Ethernet, etc.
IPv4 contains an 8-bit field called Service Type. The Service Type field is composed of a TOS (Type of Service) field and a procedure field.	The IPv6 header contains an 8-bit field called the Traffic Class Field. This field allows the traffic source to identify the desired delivery priority of its packets
The IPv4 node has only Stateful auto-configuration.	The IPv6 node has both a stateful and a stateless address autoconfiguration mechanism.
Security in IPv4 networks is limited to tunneling between two networks	IPv6 has been designed to satisfy the growing and expanded need for network security.
Source and destination addresses are 32 bits (4 bytes) in length.	Source and destination addresses are 128 bits (16 bytes) in length.
IPsec support is optional.	IPsec support is required
No identification of packet flow for QoS handling by routers is present within the IPv4 header.	Packet flow identification for QoS handling by routers is included in the IPv6 header using the Flow Label field.
Address Resolution Protocol (ARP) uses broadcast ARP Request frames to resolve an IPv4 address to a link layer address.	ARP Request frames are replaced with multicast Neighbour Solicitation messages.
Must be configured either manually or through DHCP.	Does not require manual configuration or DHCP.
ICMP Router Discovery is used to determine the IPv4 address of the best default gateway and is optional	ICMP Router Discovery is replaced with ICMPv6 Router Solicitation and Router Advertisement messages and is required.
Header includes options	All optional data is moved to IPv6 extension headers.

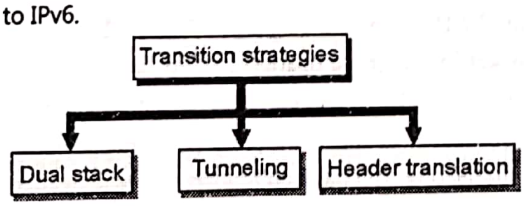
Sr. No.	BOOTP	DHCP
1.	BOOTP is designed for static environment.	DHCP is designed for dynamic environment.
2.	A manager creates a BOOTP configuration file that specifies a set of BOOTP parameters for each host.	DHCP does not require an administration to add an entry for each computer, so the database that server uses.
3.	BOOTP does not manage clients and network automatically.	DHCP was designed with aim to manage network and the clients automatically.
4.	BOOTP does not include way to dynamically assign values to individual machines.	With DHCP, network and client information need not be configured manually for individual hosts.

2.6 Transition from IPv4 to IPv6 :

- It is required to use a new version of the IP protocol. For that transition from IPv4 to IPv6 we have to define a transition day on that day each and every router or host stop using old version and should start using the new version.
- As there are huge number of systems in the Internet, transition from IPv4 to IPv6 is not practical suddenly.
- It will take some amount of time to move each and every system in the internet from IPv4 to IPv6. The transition from IPv4 to IPv6 should be smooth to prevent any problems in the system.

2.6.1 Transition Strategies :

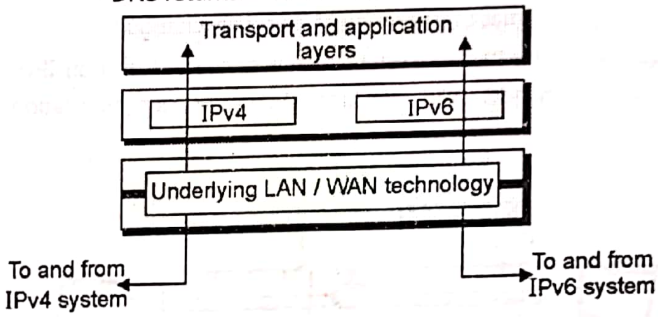
- Fig. 2.6.1(a) shows the strategies for transition from IPv4 to IPv6.



(G-2531) **Fig. 2.6.1(a) : Transition strategies**

1. Dual Stack :

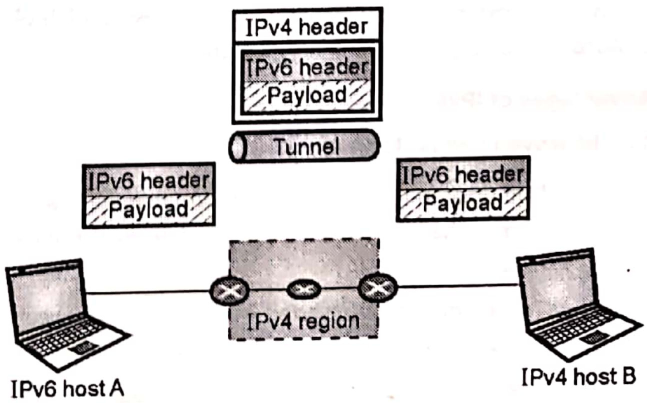
- Before completely migrating to version 6 it is recommended that all hosts should have a dual stack of protocols at the time of transition.
- Simultaneously station should run IPv4 and IPv6, until the Internet uses IPv6.
- The layout of dual stack configuration is as shown in Fig. 2.6.1(b).
- A source host send query to the DNS for deciding which version to use while sending a packet to a destination.
- A source host sends IPv4 packet if an IPv4 address is returned by the DNS, and sends IPv6 packet if DNS returns IPv6 address.



(G-2532) **Fig. 2.6.1(b) : Dual stack stategy**

2. Tunneling :

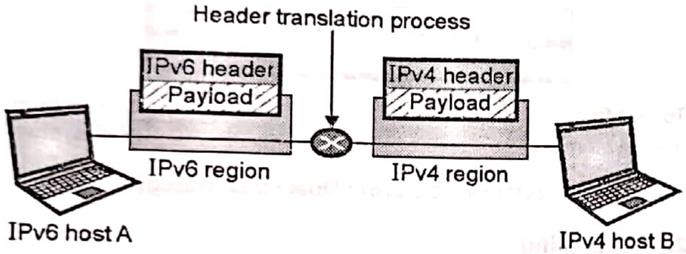
- When two computers are using IPv6 want to communicate with each other and a region through which the packet must pass uses IPv4, in such case **tunneling** strategy is used.
- The packet should have IPv4 address while passing through this region.
- When it enters in this region the IPv4 packet is encapsulated in IPv4 packet and when it exists the region it leaves its capsule.
- It looks like as if the IPv6 packet enters in a tunnel from one end and comes out from the other end.
- The protocol value is set to 41 for making it clear that IPv4 packet is holding an IPv6 packet as a data.
- The tunneling strategy is as shown in Fig. 2.6.1(c).



(G-2533) **Fig. 2.6.1(c) : Tunneling Strategy**

3. **Header translation :**

- If some systems use IPv4 and the majority of the Internet has moved from IPv4 to IPv6, in that case header translation strategy is used where the receiver does not understand IPv6 but the sender wants to use IPv6 only.
- In this situation tunneling will not work because the packet should be in the IPv4 format which has to be understood by the receiver.
- In this strategy through header translation the format of header must be totally changed.
- The IPv6 packet header is converted into an IPv4 header. Fig. 2.6.1(d) shows header translation strategy.



(G-2534) Fig. 2.6.1(d) : Header translation strategy

2.6.2 Use of IP Addresses :

- A host may need to use both IPv4 and IPv6 addresses during the transition. IPv4 addresses must disappear after completion of transition.
- During the transition it is necessary that the DNS server is to be ready to map a host name to address type.
- After migrating all hosts in the world the IPv4 dictionary will disappear.

2.7 IPv6 (Next Generation IP) :

IPv6 is the next generation Internet Protocol designed as the next step of the IP version 4. IPv6 was designed to enable high-performance and larger address space. This was achieved by overcoming many of the weaknesses of IPv4 protocol and by adding several new features.

Advantages of IPv6 :

1. **Improved header format :**

   - IPv6 uses an improved header format. In its header format the options are separated from the base header.
   - These options are inserted when needed, between the base header and upper layer data.
   - The routing process is simplified due to this modification. The speed of the routing process increases and the routing time is reduced.

2. **Larger address space :**

   IPv6 has 128-bit address, which is 4 times wider in bits is compared to IPv4's 32-bit address space. So there is a large increase in the address space.

   Address space of IPv6 = (2^{128})

3. **New options :**

   IPv6 has increased functionality due to the addition of entirely new options that are absent in IPv4.

4. **More security :**

   IPv6 includes security in the basic specification. It includes encryption of packets (ESP: Encapsulated Security Payload) and authentication of the sender of packets (AH : Authentication Header) for enhancing the security.

5. **Possibility of extension :**

   The design of IPv6 is done in such a way that there is a possibility of extension of protocol if required.

6. **Support to resource allocation :**

   To implement better support for real time traffic (such as video conference), IPv6 includes flow label in the specification. With flow label mechanism, routers can recognize to which end-to-end flow the given packet belongs to.

7. **Plug and play :**

   IPv6 includes plug and play in the standard specification. It therefore must be easier for novice users to connect their machines to the network, it will be done automatically.

8. **Clearer specification and optimization :**

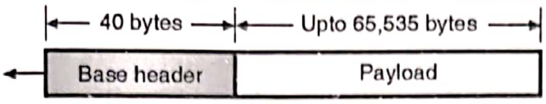
   IPv6 follows good practices of IPv4, and omits flaws/obsolete items of IPv4.

2.7.1 IPv6 Packet Format :

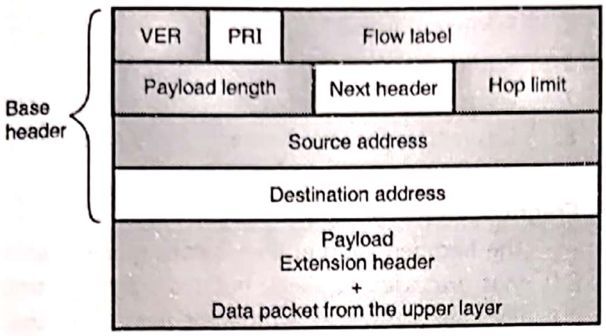
- Fig. 2.7.1(a) shows IPv6 packet. Fig. 2.7.1(b) shows the packet format (Base header) of IPv6. Each packet can be divided into two parts viz : base header and payload.
- Base header is the mandatory part and payload is an optional one. The payload follows the base header.

  The payload is made up of two parts :

  1. An optional extension headers and
  2. The upper layer data.
- The base header is 40 byte long whereas the payload consisting of the extension header and upper layer data can have information worth upto 65, 535 bytes.



(G-2245)**Fig. 2.7.1(a) : IPv6 packet**



(G-550) **Fig. 2.7.1(b) : Format of an IPv6 datagram (Base header)**

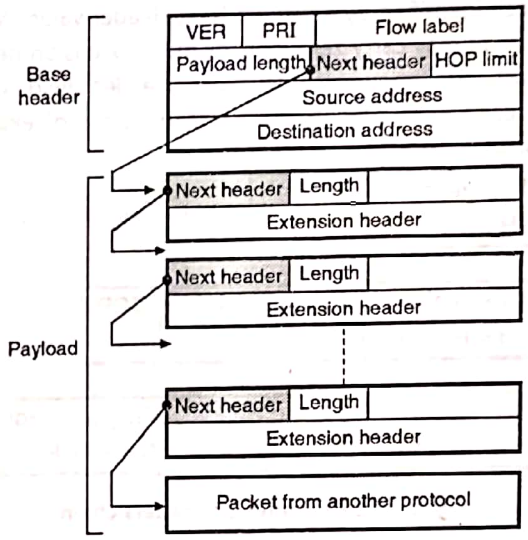
Base header :

Fig. 2.7.1(b) shows the base header. It has eight fields. These fields are as follows :

1. **Version (VER) :** The contents of this 4 bit field defines the version of IP such as IPv4 or IPv6. If VER = 6, then the version is IPv6.
2. **Priority :** This 4 bit field contents defines the priority of the packet which is important in connection with the traffic congestion.
3. **Flow label :** It is a 24 bit (3 byte) field which is supposed to provide a special handling for a particular flow of data.
4. **Payload length :** The contents of the 16 bit or 2 byte length field are used to indicate the total length of the IP datagram excluding the base header. That means it gives the length of only the payload part of the datagram.
5. **Next header :** It is an 8 bit field which defines the header which follows the base header in the datagram.
6. **Hop limit :** Contents of this 8 bit (1 byte) field have the same function as TTL (time to live) in IPv4.
7. **Source address :** It is a 16 byte (128 bit) Internet address which corresponds to the originator or source which has produced the datagram.
8. **Destination address :** This is a 16 byte (128 bit) internet address which corresponds to the address of the final destination of datagram. But this field will contain the address of the next router and not the final destination if source routing is being used.

2.7.2 Payload :

- The meaning and format of payload field in IPv6 is different as compared to payload field in IPv4.
- Fig. 2.7.2 shows payload field in IPv6.
- In IPv6, the payload is combination of zero or more extension headers (options) which is followed by data from other protocols such as UDP, TCP etc.
- In IPv4, option is a part of the header, whereas in IPv6 it is designed as extension headers.
- Depends on the situation the payload can have as many extension headers as required.
- Extension header is made up of two mandatory fields : next header and the length which is followed by information which is related to the particular option.
- Value of next header field i.e. code defines which type of the next header is (e.g. source routing options, fragmentation option etc.)
- The last next header describes the protocol which carries the datagram.
- Some next header codes are listed in Table 2.7.1.



(G-2246) **Fig. 2.7.2 : IPv6 payload**

Table 2.7.1 : Next header codes

Sr. No.	Code	Next header code
1.	00	HOP by hop option
2.	02	ICMPv6
3.	06	TCP
4.	17	UDP
5.	43	Source routing option
6.	44	Fragmentation option
7.	50	Encrypted security payload
8.	51	Authentication header
9.	59	Null (no next header)
10.	60	Destination option

2.8 Extension Headers :

- As stated earlier the length of the base header is 40 bytes and it always remains constant.
- But in IPv6, the fixed base header can be followed by upto six extension headers. In IPv4 these are optional headers.
- This gives more functionality to the IP datagram.
- The IPv4 header has space for some optional fields requiring a particular processing of packets. These optional fields are not used often, and they can deteriorate router performance because their presence must be checked for each packet. IPv6 replaces these optional fields by **extension headers.**
- In IPv6, optional Internet-layer information is encoded in separate headers that may be placed between the IPv6 header and the upper-layer header in a packet (see Fig. 2.8.1).
- There are a small number of such extension headers, each identified by a distinct Next Header value. An IPv6 packet may carry zero, one, or more extension headers, each identified by the Next Header field of the preceding header. There are seven kinds of extension header :

IPv6 Header NH = TCP	TCP

NH = Next Header

IPv6 Header NH = Routing	Routing Header NH = TCP	TCP

IPv6 Header NH = Routing	Routing Header NH = Frag. Hdr	Fragment Hdr NH = TCP	TCP

Fig. 2.8.1 : Examples of headers chain

- Extension headers are not examined or processed by any node along a packet's delivery path, until the packet reaches the node (or each of the set of nodes, in the case of multicast) identified in the Destination Address field of the IPv6 header, except for the Hop-by-Hop Options header and the Routing header.
- Therefore, extension headers must be processed strictly in the order of their appearance in the packet; a receiver must not, for example, scan through a packet looking for a particular kind of extension header and process that header before processing all the preceeding ones.
- Each extension header has a length equal to a multiple of 64 bits (8 bytes). A full implementation of IPv6 must include support for the following extension headers :
- When more than one extension header is used in the same packet, it is recommended that those headers appear in the following order :

1. IPv6 header
2. Hop-by-Hop Options header
3. Destination Options header
4. Routing header
5. Fragment header
6. Authentication header
7. Encapsulating Security Payload header
8. Destination Options header
9. Upper-layer header

1. Fragmentation :

- The fragmentation in IPv6 is conceptually same as that discussed for IPv4, but the fragmentation in IPv6 takes place at a different place than that in IPv4.
- In IPv4 the fragmentation is done by the source or router, but in IPv6 the fragmentation may be carried out only by the original source.

2. Authentication and Privacy :

IPv6 provides authentication and privacy using options in the extension header.

Review Questions

Q. 1 Compare IPv4 and IPv6.

Q. 2 What is fragmentation ? Explain how is it supported in IPv4 and IPv6.

Q. 3 Explain the addressing scheme in IPv4 and IPv6. When IPv6 protocol is introduced, does the ARP protocol have to be changed ? Explain.

Q. 4 What is fragmentation ? Explain how it is supported in IPv4 and IPv6.

Q. 5 Write short note on ICMPv6.

Q. 6 State and explain various notations used for IPv6 addresses.

Q. 7 Explain the CIDR notation.

Q. 8 What is the address space of IPv6 ?

Q. 9 Explain the three address types defined by IPv6.

Q. 10 Explain the following for IPv6.

1. Loopback address.
2. Compatible address.

Q. 11 Explain the following feature of IPv6 : Autoconfiguration.

Q. 12 Write a note on : Transition from IPv4 to IPv6.

Q. 13 What is tunneling ? How is it used in transition from IPv4 to IPv6.

Q. 14 State and explain any four advantages of IPv6.

Q. 15 Explain the IPv6 packet format.

Q. 16 Write a note on : extension headers in IPv6.